

PURBANCHAL UNIVERSITY

2024

Master of Computer Application (M.C.A.)/First Semester/Final

Time: 03:00 hrs.

Full Marks: 60 /Pass Marks: 24

MCA104/MCA114: Advanced Database Management System (New/Old)

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group A

Answer TWO questions.

2×12=24

1. Check whether the schedule is conflict serializable or not. Also find out equivalent serial schedule.

S: R2(A); R1(B); W2(A); R3(A); W2(B); W3(A); R2(B); W2(B)

2. Consider the schema for Company database:

EMPLOYEE(SSN, Name, Address, Sex, Salary, SuperSSN, DNO)

DEPARTMENT(DNo, DName, MgrSSN, MgrStartDate)

DLOCATION(DNo, DLoc)

PROJECT(PNo, PName, PLocation, DNo)

WORKS_ON(SSN, PNo, Hours)

Write SQL queries to

- (a) Make a list of all project numbers for projects that involve an employee whose last name is 'Scott', either as a worker or as a manager of the department that controls the project. 2
- (b) Show the resulting salaries if every employee working on the 'IoT' project is given a 10 percent raise. 2
- (c) Find the sum of the salaries of all employees of the 'Accounts' department, as well as the maximum salary, the minimum salary, and the average salary in this department. 4
- (d) Retrieve the name of each employee who works on all the projects controlled by department number 5 (use NOT EXISTS operator). 2
- (e) For each department that has more than five employees, retrieve the department number and the number of its employees who are making more than Rs. 6,00,000. 2

Contd. ...

(2)

3. What is log-based recovery technique? Explain immediate update and deferred update techniques with suitable example.

Group B

Answer SIX questions.

6×6=36

4. What is RAID-5 and how does it work? How does RAID improve data redundancy and fault tolerance?
5. Explain architecture of DBMS with a suitable example.
6. What is XQuery, and how is it used to query XML data? What is a schema, and how is it used in semi-structured data?
7. What is the difference between a clustered index and a non-clustered index? How does indexing improve search and retrieval operations in DBMS?
8. What are the emerging database technologies, and how do they differ from traditional database technologies in terms of their features and capabilities?
9. What is rule-based query optimization, and how does it work in database system?
10. Differentiate between authentication and authorization with respect to database security. Explain various levels of database security.

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